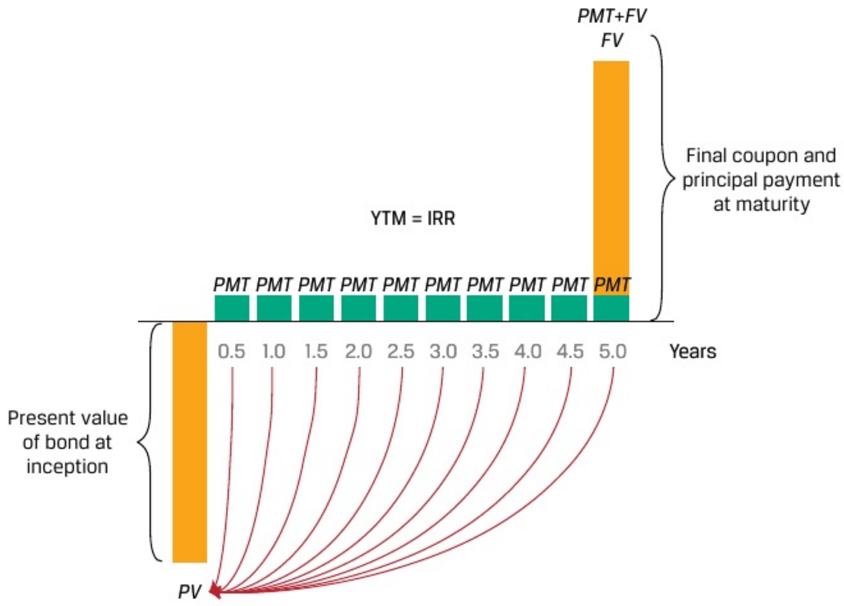
Exhibit 8: Coupon Bond Implied Return



The YTM assumes an investor expects to receive all promised cash flows through maturity and reinvest any cash received at the same YTM. For coupon bonds, this involves periodic interest payments only, while for level payment instruments such as mortgages, the calculation assumes both interest and amortized principal may be invested at the same rate. Like other internal rates of return, the YTM cannot be solved using an equation, but it may be calculated using iteration with a spreadsheet or calculator, a process that solves for r in Equation 19, as follows:

$$PV(Coupon\ Bond) = PMT_1 / (1 + r)^1 + PMT_2 / (1 + r)^2 + \ldots + (PMT_N + FV_N) / (1 + r)^N, \quad (19)$$

where FV equals a bond's principal and N is the number of periods to maturity.

The Microsoft Excel or Google Sheets YIELD function can be used to calculate YTM for fixed-income instruments with periodic interest and full principal payment at maturity:

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= YIELD (settlement, maturity, rate, pr, redemption,
frequency, [basis])
```

where:

settlement = settlement date entered using the DATE function;

maturity = maturity date entered using the DATE function;

rate = semi-annual (or periodic) coupon;

pr = price per 100 face value;

redemption = future value at maturity;

frequency = number of coupons per year; and

[basis] = day count convention, typically 0 for US bonds (30/360 day count).

Example 9 illustrates the implied return on fixed income instruments with periodic interest.